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Reading IMF Programmes on Day One

Predicting programme interruptions with approval-time information

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INTRODUCTION AND LITERATURE REVIEW

Introduction and financial relevance

The International Monetary Fund (IMF) supports countries facing balance-of-payments difficulties through lending programmes that combine financing with policy commitments and periodic reviews. Approval does not guarantee that the agreed programme will be implemented. We ask whether information available at Board approval can predict whether a programme is later interrupted (see 1b for the outcome definition).

Interruption matters financially because implementation can affect expected financing and confidence in the government's policy commitments. Mody and Saravia (2006, pp. 843-845, 857) find that IMF programmes do not uniformly improve access to international bond markets; favourable effects depend on credible reform commitments and economic circumstances. Reinsberg et al. (2022, pp. 1023, 1034–1035) associate permanent interruptions with lower institutional investor ratings and a short-lived increase in sovereign bond spreads. These findings motivate studying interruption risk.

Political incentives and implementation capacity

The implementation literature emphasises the borrowing government's willingness and ability to fulfil its commitments. Ivanova et al. (2003, pp. 4, 34) identify political cohesion, stability and bureaucratic quality as important influences on implementation. Joyce (2006, pp. 342–345, 351) similarly links implementation to government ideological cohesion, regime duration and political and economic openness. Dreher (2003) shows that electoral incentives interact with political institutions and also identifies associations with initial economic vulnerability. Together, these studies suggest that implementation depends on domestic constraints as well as the agreement with the IMF.

Our 13 inputs do not directly measure elections, political cohesion or bureaucratic quality, so these mechanisms are only partly observed. Programme characteristics may capture some of these borrower differences indirectly.

Conditionality and programme design

Evidence on conditionality is mixed. Ivanova et al. (2003, pp. 4, 34) find limited explanatory power for conditionality once domestic political factors are considered. Dreher's (2009, pp. 248–250) survey discusses competing arguments about conditions as commitment devices and constraints on implementation. More recently, Reinsberg et al. (2022, pp. 1023, 1026, 1035) find that greater structural conditionality at programme onset is associated with interruption, especially conditions involving public-sector reform, privatisation and price liberalisation.

We therefore test the initial number of structural conditions and prior actions. Prior actions are measures required before approval, while structural conditions concern reform commitments during the programme. Financing relative to quota, planned duration and scheduled reviews describe further aspects of the arrangement. Concessional and extended-facility indicators distinguish programme types and may capture both design and borrower characteristics.

The outcome definition also affects interpretation: a programme with more scheduled reviews has more chances to leave one uncompleted, so duration and review counts may predict interruption partly through exposure. Counting conditions also misses their political difficulty and content.

Macroeconomic vulnerability and previous IMF use

Weak growth, inflation and external financing pressures can make agreed adjustment harder to implement. Iseringhausen et al. (2019, pp. 8, 16, 19, 23–25) find that macroeconomic conditions, institutions and previous IMF experience help explain repeated programme use. Their outcome, however, is a successor arrangement within three years. A country can return to the IMF after successfully completing a programme because it experiences a new shock. Evidence on repeated use therefore suggests candidate predictors, which we test against interruption.

Our model tests growth, inflation and the current-account balance, using IMF staff estimates or projections recorded at approval. It also includes the number of earlier programmes, time since the preceding programme's planned end and an indicator for no previous track record. These variables summarise economic circumstances and programme history; reserves and direct institutional indicators are absent. We use the approval snapshot because later revisions or realised outcomes would give the model information unavailable at the time of prediction.

Machine learning and the research contribution

Machine learning already contributes to IMF forecasting. Agbloyor et al. (2023, pp. 1-5) predict IMF bailouts using 6,550 observations and 138 features, reporting predictive information from economic, agricultural, energy, health and social indicators. Batsuuri et al. (2024, pp. 12, 14–18, 22–25) find that machine-learning methods outperform traditional approaches when forecasting new IMF arrangements, with results sensitive to preprocessing

and prediction horizon. Both studies predict participation; we predict what happens after a programme is approved.

Gu et al. (2020, pp. 2223, 2225–2227), in The Review of Financial Studies, demonstrate the value of nonlinearities and interactions in equity-return prediction. This motivates comparing flexible models, although our sample of 173 programmes is far smaller and our target different. Breiman (2001, pp. 5–8, 10–11) provides the basis for random forests, which combine predictions from randomised trees. We compare the random forest with logistic regression, an interpretable benchmark for a binary outcome, and a neural network. The comparison tests whether additional flexibility improves prediction in a small sample.

Our contribution is a focused predictive test in the sense of Shmueli (2010), discussed in 1b: can the selected approval-time variables classify interruption, and does a random forest improve on logistic regression and a neural network? The literature covers implementation mechanisms, financial consequences and predictors of participation or repeated use, but has not tested this framework.

We use country-grouped cross-validation within the training sample to tune the random forest and the neural network, and evaluate all models on later programmes approved during 2014–2016. The two checks test different kinds of generalisation: to unseen countries during validation, and to later approvals, which may include countries already in the training data. We report balanced accuracy alongside overall accuracy (see 1b). Coefficients and permutation importance are interpreted as predictive associations, not as causal effects of changing programme characteristics.

STUDY GOAL

We test whether approval-day information can predict if an International Monetary Fund (IMF) programme will be interrupted. The unit of observation is one lending programme. The prediction is made at Board approval, using only data known on that date.

This is a supervised binary classification task. The output, interrupted, equals 1 when fewer reviews were completed than scheduled at approval. It equals 0 when every scheduled review was completed. The inputs are 13 variables in four groups. These are deal design, conditionality, IMF track record, and macroeconomic conditions. Section 1g lists them.

Following James et al. (2013, pp. 16-17), we estimate f in $Pr(Y = 1 \mid X) = f(X)$. The estimate is judged on programmes it has not seen during training. Following Shmueli (2010, p. 291), we treat this as a prediction problem. Coefficients and variable importance are not read as causal effects.

Performance is measured on a later test set. We report accuracy and balanced accuracy, the mean of the hit rates on both classes. Interruptions are the majority in the test set (see 1d). A rule that always predicts interruption therefore gets high accuracy. Its balanced accuracy is only .50. Plain accuracy misleads when classes are unequal (He & Garcia, 2009, p. 1276). Balanced accuracy removes this advantage. Models are tuned on balanced accuracy for the same reason.

The IMF pays its financing in tranches. Each tranche after the first requires the Board to complete a scheduled review (IMF, n.d.). The review checks the quantitative targets and structural conditions agreed at approval. When a review is not completed, the next disbursement is withheld. The country then loses financing the programme had scheduled. An approval-day flag could inform programme design at the IMF and risk assessment by creditors.

STUDY DESIGN AND DATA COLLECTION

The study is a cross-section of IMF programmes. Each row is one programme, described as it stood at approval. We keep facilities with scheduled reviews, including their predecessors. These are the Stand-By Arrangement (SBA), the Extended Fund Facility (EFF), the Extended Credit Facility (ECF), and the Standby Credit Facility (SCF). Facilities without reviews and precautionary arrangements are dropped (see 1d). Approvals run from 2002 to 2016. Review records in the Monitoring of Fund Arrangements (MONA) database start in 2002. Ending the window in 2016 limits overlap with the COVID-19 period.

No single public source records programme design and review outcomes together. We therefore built the dataset from IMF files. The IMF Financial Data Query Tool gives the history of lending arrangements (IMF, 2026a). It provides dates, facility type, and access in percent of quota at approval. MONA covers reviews, conditions, and staff macro estimates (IMF, 2026b). We use four MONA files. Reviews lists scheduled and completed reviews, from which we build the output. Combined lists structural conditions and prior actions. Mecon holds staff estimates of growth, inflation, and the current account. Description holds a cancellation flag, used only to check the output. We also downloaded the April 2026 World Economic Outlook (WEO; IMF, 2026c). We also used the World Bank's World Development Indicators (WDI; World Bank, 2026). The final macro inputs come from MONA; Section 1g explains why.

The commitments file and MONA share no common identifier. Country names are spelled differently, and approval dates can differ by a few days. We normalise country names and match on the closest approval date within 60 days. Blends approved for the same country on the same day become one programme. They share one review schedule. Section 1d reports the resulting sample.

We split the sample by approval year. Training covers 2002 to 2013, and the test set covers 2014 to 2016. The test programmes are never used to fit or tune the models. All inputs are taken as they stood on approval day (see 1g). Hyperparameters are tuned on the training set with 10-fold cross-validation and leave-one-out cross-validation (LOOCV). A third scheme holds out whole countries in each fold. Many countries have several programmes, so random folds put one country on both sides. We use the country-grouped scores to choose the tuned settings for the random forest and the neural network.

DATA AND VARIABLES

Data and summary statistics

Our final dataset covers 173 IMF lending programmes in 86 countries, approved between 2002 and 2016; 82 of them (47.4%) were interrupted.

Starting from the IMF's list of 1,576 lending arrangements since 1952, we keep the 221 non-precautionary arrangements approved in 2002–2016 and drop rapid one-off loans (RCF, RFI) and the Precautionary and Liquidity Line, which have no review schedule and so cannot be interrupted. Merging blended programmes into one observation leaves 182 programmes, of which 173 match the IMF's Monitoring of Fund Arrangements (MONA) database. A programme counts as interrupted if fewer reviews were completed than were scheduled at approval. We split by approval year (training 2002–2013, n = 145; test 2014–2016, n = 28) and fill the few missing values with training-set medians, so nothing from the test period leaks into training.

Table 1. Summary statistics of the 13 inputs (before imputation, n = 173)

Variable	Unit	Missing	Mean	SD	Min	Median	Max
log_access_pct_quota	log of % of quota	0	4.62	1.29	1.61	4.50	8.08
planned_months	months	0	32.58	7.56	7.20	35.98	47.97
n_reviews_sched	count	0	6.60	2.27	2	6	16
n_struct_cond	count	1	10.67	4.98	2	10	26
n_prior_actions	count	1	2.79	2.98	0	2	16
prev_n_20y	count	0	3.35	2.06	0	3	8
years_since_prev	years	11	4.09	6.82	0	1.62	47.74
growth	%	5	4.04	4.23	-8.69	4.15	24.46
inflation	%	8	7.73	7.93	-17.69	5.50	48.04
current_account_gdp	% of GDP	10	-8.62	11.77	-92.49	-6.20	15.36
concessional	0/1	0	0.64	0.48	0	1	1
extended	0/1	0	0.69	0.46	0	1	1
no_track_record	0/1	0	0.06	0.25	0	0	1

For binary variables the mean is the share of programmes with value 1. Source: own calculations, summary_statistics.csv.

The classes are close to balanced overall, but interruptions rise from 39% of approvals in 2002–2004 to 61% in 2014–2016. They are the minority in training (65 of 145, 44.8%) and the majority in the test set (17 of 28, 60.7%). Always predicting "interrupted" would therefore already be right for 60.7% of test programmes, which is the benchmark the models in Task 2 have to beat.

ECFs, concessional loans to low-income countries, make up 97 of the 173 programmes, followed by 47 SBAs, 16 EFFs, 8 blends and 5 SCFs. Interruption rates range from 64% for SBAs to 38% for ECFs and blends.

A typical programme runs for three years with six scheduled reviews, ten structural conditions and two prior actions. Access in percent of quota is strongly right-skewed, so we use

its log, and the macro variables have long tails (inflation runs from -18% to 48%). Missing values are rare: years_since_prev is undefined for the 11 first-time borrowers, and no other input misses more than 10 values.

Interrupted programmes differ from completed ones mainly in deal structure. They schedule more reviews (7.3 against 5.9), receive more access (mean log access 4.96 against 4.31) and are less often concessional (52% against 74%) or extended (59% against 79%), all with t-test p-values below 0.01. Gaps in conditionality, track record and macro conditions are smaller, with p-values of 0.09 or more.

The test period also differs in its inputs: its programmes are larger, longer and carry more conditions, and only half are concessional (against 66% in training). Test accuracy therefore also shows how well a model transfers to a later period, not just how well it fits.

Choice of input and output variables

The output is whether a programme was interrupted. All 13 inputs are known on the day the IMF Executive Board approves the programme, which is when the prediction is made.

Selection principle. A predictive input must be available at the time of prediction and carry information about the outcome; it needs no causal interpretation (Shmueli, 2010). We therefore use only approval-day information: the "upon approval" columns of the commitments list and MONA's approval snapshot (R0).

Output. Tying the label to the approval-day review schedule means later rescheduling cannot change it. MONA's cancellation flag would be too narrow: only 29 of the 82 interrupted programmes were formally cancelled, while the other 53 ended with reviews outstanding.

Inputs. Table 2 groups the inputs into deal structure, conditionality, track record and macroeconomic conditions.

Table 2. Input and output variables

Variable	Group	Description	Source	Why it should help predict
interrupted	Output	1 = fewer reviews completed than scheduled at approval	MONA Reviews	The event to predict
log_access_pct_quota	Deal structure	Log of agreed access, % of IMF quota	Commitments, upon approval	Larger programmes signal larger problems and raise the stakes at each review
planned_months	Deal structure	Planned length, approval to scheduled expiry	Commitments, upon approval	Longer programmes are exposed longer to shocks and changes of government
concessional	Deal structure	1 = includes a concessional (PRGT) facility	Commitments	Low-income borrowers face different circumstances and incentives
extended	Deal structure	1 = EFF or ECF	Commitments	Medium-term structural programmes vs. short-term crisis programmes
n_reviews_sched	Deal structure	Reviews scheduled at approval	MONA Reviews, R0	Each review is a point at which the programme can stop
n_struct_cond	Conditionality	Structural conditions at approval	MONA Combined, R0	More reform conditions are harder to meet on time
n_prior_actions	Conditionality	Prior actions before Board approval	MONA Combined, R0	Signals low trust, or commitment already shown
prev_n_20y	Track record	Programmes in the 20 years before approval	Commitments history	Repeated users may have persistent implementation problems
years_since_prev	Track record	Years since the previous programme's planned end (0 if overlapping)	Commitments history	A quick return suggests the previous programme did not solve the problem
no_track_record	Track record	1 = no earlier programme	Commitments history	First-time borrowers have no history with the IMF

Table 2 (continued)

Variable	Group	Description	Source	Why it should help predict
growth	Macro	Real GDP growth, %, staff estimate for the approval year	MONA Mecon, R0	A weak economy makes programme targets harder to reach
inflation	Macro	CPI inflation, %, staff estimate for the approval year	MONA Mecon, R0	High inflation signals macroeconomic instability
current_account_gdp	Macro	Current-account balance, % of GDP	MONA Mecon, R0	The size of the external gap the programme must close

Transformations and overlap. Access enters in logs. Facility type is left out because concessional and extended identify it exactly, and the number of tranches because it is almost always the number of reviews plus one. Some inputs remain correlated (extended with planned_months, r = 0.77), which does not harm a random forest's predictions.

Excluded variables. Everything known only after approval is excluded: amounts drawn, completed reviews, actual end dates and later MONA snapshots. Fiscal variables are left out because the WEO has only the revised 2026 data and MONA lacks them for 24–45% of programmes. Identifiers such as country and approval date are excluded too, since the model should generalise to new programmes. The full list is in the excluded_variables sheet of the Excel file.

Caveats. The number of scheduled reviews enters both an input and the outcome definition, so part of its predictive power is mechanical. The macro inputs are staff projections at approval, which may be optimistic but are what was known at the time.

RANDOM FOREST

The first model that is used is a random forest. As our study is a prediction through classification, using random forest first appeared as being a promising tool (Batsuuri et al., 2024, p. 3). This model combines many classification trees, each splitting the data through a series of simple yes/no questions.

A single tree captures nonlinear patterns and interactions, but it is unstable: small changes in the data can produce a very different tree. For this reason, we use bagging (bootstrap aggregation). The idea is to grow many trees, each on a bootstrap sample (programs drawn with replacement), and average their votes. Averaging helps to cancel out noisy predictions. However, bagging alone has a weakness. If one feature is very strong, almost every tree will split on it first, making the trees similar, which limits the gain from averaging (James et al., 2013, p. 320). Indeed, Hastie et al. (2009, p. 588) formalized it with the variance of an average of B trees, each with variance σ^2 and pairwise correlation ρ:

$$\mathbf{Var = \rho\sigma^2 + (1 - \rho)\sigma^2 / B}$$

More trees (B) shrink the second term, but the first term only shrinks with less correlated trees. We use Breiman's (2001, p. 6) solution of limiting the features available at each split to a random subset of *mtry* inputs. Each tree then votes: the share of trees voting "interrupted" gives the predicted probability, and a threshold assigns the class. The programs left out of each bootstrap sample (out-of-bag) provide a built-in error estimate.

The default random forest was trained with the *randomForest* R library default hyperparameters: 500 trees, *mtry* 3. The OOB balanced accuracy scored 0.609.

To tune our hyperparameters (number of trees and *mtry*), we scored 24 different pairs on balanced accuracy, ranging from 100 to 1000 trees and *mtry* from 2 to 12 with the caret package. Balanced accuracy is the average of two hit rates: the share of interrupted programs the model catches and the share of completed programs it catches. A model that always predicts the same outcome scores 0.50, whatever the class shares.

We used three different cross-validation schemes: 10-fold CV, LOOCV, and 10-fold CV by country. The last is our main measurement, as it keeps all programs of a country in the same fold, so the forest is never scored on a country it was trained on.

Balanced accuracy served as our metric of choice because the classes are unequal and shift between development and test sets. For 10-fold CV and country CV, we compute it within each fold and average over the ten folds; for LOOCV, where each fold holds a single program, we compute it once over all out-of-fold predictions.

Validation	Default forest	Tuned pair	Tuned forest
Out-of-bag (default)	0.609	-	-
10-fold CV	0.623	250 trees, mtry 2	0.657
10-fold CV by country	0.597	250 trees, mtry 3	0.620
LOOCV	0.606	100 trees, mtry 8	0.635

Table 3. Default against tuned forest, development set (balanced accuracy)

The default pair (500, 3) is part of the tuning grid, so its CV scores come from the same runs. Out-of-bag accuracy of the default forest is 0.628.

Country CV gives the lowest scores (best 0.620, against 0.657 for 10-fold CV and 0.635 for LOOCV), as expected, since the forest is never tested on a country it has seen. We still select it because programs of the same country share many features, so scoring the forest on countries it was trained on would flatter the result; country CV gives the most conservative estimate.

We keep the pair chosen under country CV (250 trees, mtry 3), refit it on all 145 development programs with the threshold that maximises out-of-fold balanced accuracy (0.51), and predict the 28 test programs once.

Model	Accuracy	Sensitivity	Specificity	Balanced accuracy	AUC
Always “interrupted”	0.607	1.000	0.000	0.500	0.500
Default forest (threshold 0.50)	0.571	0.588	0.545	0.567	0.620
Final forest: 250 trees, mtry 3 (threshold 0.51)	0.500	0.471	0.545	0.508	0.586
Final forest, dev OOF	0.641	0.523	0.738	0.630	0.631
Final forest threshold 0.50 (for comparison with other models)[1]	0.571	0.588	0.545	0.567	0.586

Table 4. Final model and predictions, test set (28 programs, 2014 to 2016)

On the 28 test programs, the final forest reaches a balanced accuracy of 0.508, barely above the 0.500 of always predicting “interrupted”, and its accuracy (0.500) is even below that rule’s (0.607), which shows why plain accuracy misleads when classes are unequal. Out of fold, it reaches 0.630; this is more precise than the test score but slightly optimistic, since both the pair and the threshold were chosen on these predictions. It also differs from Table 3’s 0.620 because it pools all predictions at the chosen threshold (0.51) instead of averaging folds at 0.50. The default forest scores 0.567 on test, but the gap comes entirely from two interrupted programs with a vote share of 0.504, classified as completed at 0.51: at 0.50, the final forest scores the same 0.567. On the test set, one extra correct prediction moves balanced accuracy by 0.03 to 0.05. The forest thus finds a modest signal in development, but no reliable advantage over chance on later programs.

Random forests are considered black-box models because hundreds of trees cannot be read directly. However, predictor contribution is still identifiable, as explained in the section detailing permutation importance.

Permutation importance

This method shuffles the values of each predictor among the out-of-bag samples, breaking its link with the target, and measures the resulting drop in accuracy, averaged over all trees. Higher values mean higher predictive power, while small or negative values signal no contribution. The mean decrease in accuracy (MDA) is one of the two importance measures in randomForest, the other being the mean decrease in Gini impurity (Abedin et al., 2021, p. 23). Importance indicates neither direction nor causality, and correlated predictors (*extended* and *concessional*) share credit.

[1]The 0.50 row uses the same threshold as the other models, for a like-for-like comparison; it differs from 0.51 only because two interrupted programs sit at a vote share of 0.504.

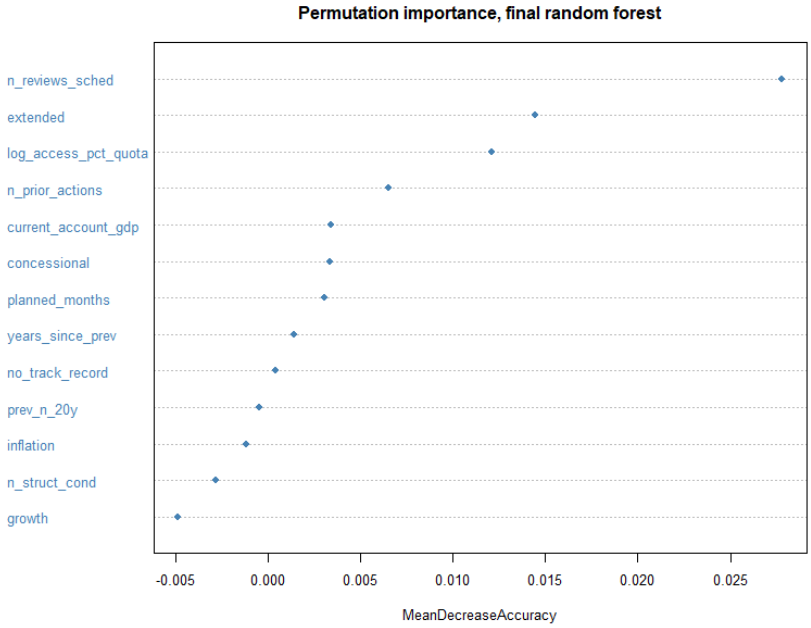


Figure X: Permutation Importance

The number of scheduled reviews is the strongest predictor of interruption: 85% of development programs with eight or more reviews were interrupted, against 36% of the others. Part of this is mechanical, as each extra review is one more chance to miss one. Program length itself predicts little, even though Reinsberg et al. (2022, p. 1028) find that longer agreements fail more often. Extended facilities come second, with 38% of extended programs interrupted against 60% of the others, in line with the IMF's review: "EFF and EFF-ECF arrangements, generally longer in duration than SBAs, had higher completion rates" (IMF, 2019, p. 45). Loan size ranks third and prior actions fourth (48% interrupted with at least one prior action, against 37% without), in line with the IMF's finding of "a negative association between PAs at arrangement approval and completion rates" (IMF, 2019, p. 45).

Growth and inflation add no predictive value, consistent with regression studies that find initial economic conditions largely insignificant (Ivanova et al., 2003, p. 1; Reinsberg et al., 2022, p. 1028). The literature disagrees on conditionality: Reinsberg et al. (2022, p. 1028) find that structural conditions raise the risk of interruption, while Ivanova et al. (2003, p. 1) find no material effect. Our forest is closer to the latter, as the number of structural conditions adds no predictive value.

ALTERNATIVE MODEL: LOGISTIC REGRESSION

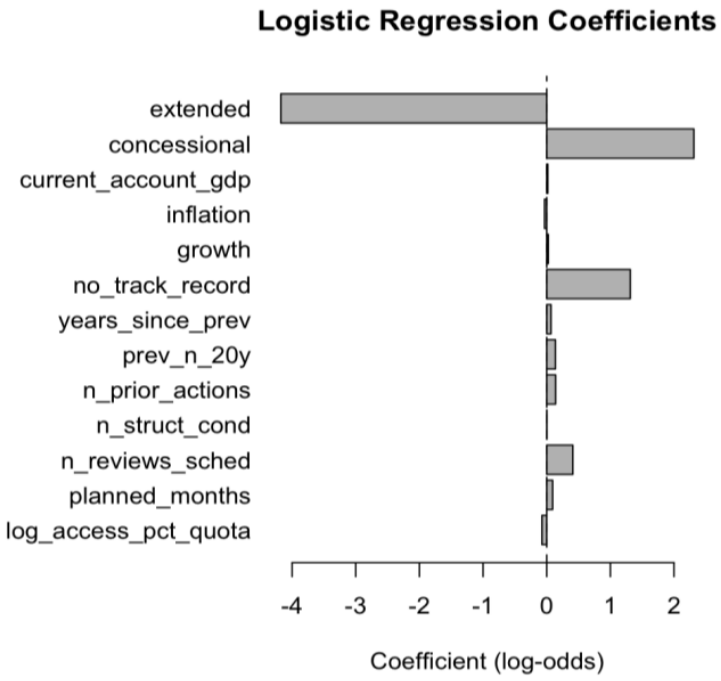
To evaluate an alternative to the random forest, we used a logistic regression model to predict whether an IMF lending program would be interrupted. We chose logistic regression because the dependent variable, *interrupted*, is binary (Hosmer et al., 2013, p. 1; James et al., 2013, pp. 130–131). It provides a relatively simple and interpretable benchmark against which the performance of a more complex model can be analyzed.

The model was trained on the same 13 explanatory variables as the random forest, covering program design, conditionality, previous program experience and macroeconomic conditions. These include economic growth, inflation, whether an arrangement is extended or concessional, and the number of prior actions required before approval. The model estimates the probability of an interruption, which is converted into a binary prediction using a threshold of 0.50 (James et al., 2013, p. 146). It was not tuned, while the random forest was tuned and its threshold searched out of fold. The same training and test sets were used for both models, allowing a direct comparison.

On the test set of 28 programs, logistic regression achieved an accuracy of 0.536, a balanced accuracy of 0.570 and an AUC of 0.684, against 0.500, 0.508 and 0.586 for the random forest at its selected threshold of 0.51. At the same 0.50 threshold, however, the forest reaches 0.571 and 0.567, so the two models differ mainly in AUC, which does not depend on the threshold (Fawcett, 2006): logistic regression ranks interrupted programs above completed ones more consistently. With 28 test programs, even this gap should be read cautiously, and both results show that predicting interruptions remains difficult.

A key advantage of logistic regression is its interpretability. Its coefficients indicate how each explanatory variable is associated with the log-odds of interruption, holding the other variables constant (Hosmer et al., 2013, pp. 49–52; see Figure 1). The largest coefficients belong to extended facilities (−4.18, $p = .004$) and concessional facilities (2.31, $p = .087$), of which only the former is statistically significant at the 5% level. These two variables are strongly correlated in the training data ($r = .81$). As a result, the model splits their joint association into opposite signs and estimates both with large standard errors, so their individual coefficients should not be read in isolation (James et al., 2013, pp. 99–102). The absence of a track record also has a sizeable positive coefficient (1.31), but it is imprecisely estimated ($p = .231$). More generally, these associations are not causal, and coefficient magnitudes are not directly comparable because the predictors are measured on different scales (Hosmer et al., 2013, pp. 62–64).

Figure 1
Logistic Regression Coefficients for Program Interruption



Note. Coefficients are in log-odds and estimated on the training set. Positive values indicate higher odds of interruption. Predictors are on different scales, so bar lengths are not directly comparable measures of importance. Extended and concessional are strongly correlated (r = .81).

The random forest, in contrast, can capture non-linear relationships and interactions between variables (Breiman, 2001a), at the cost of interpretability (Breiman, 2001b; James et al., 2013, pp. 24–26). Logistic regression imposes a simpler functional form and provides a more transparent account of the relationships it represents.

Overall, logistic regression matched the random forest on balanced accuracy and ranked programs better, while offering greater interpretability. One possible explanation is that the simpler model generalized better, whereas the random forest's flexibility may have made it more susceptible to fitting noise in the training data (Hastie et al., 2009, pp. 219–223; James et al., 2013, pp. 33–36). However, this cannot be established from the results alone, and with only 28 test programs the differences should be interpreted cautiously. The comparison shows that a more complex model does not necessarily produce better predictions (Christodoulou et al., 2019; Couronné et al., 2018), and that interpretability remains an important consideration when evaluating models for financial decision-making (Rudin, 2019).

MODEL AND SET-UP

A feedforward neural network passes the inputs through layers of neurons, each of which-h applies a non-linear activation function, such as the sigmoid, to a weighted sum of its inputs. This non-linearity is what lets the network learn non-linear relationships (Wüthrich & Merz, 2023, pp. 269-273). We use the R package nnet, which fits networks with a single hidden layer and optimises the weights with the BFGS method (Ripley & Venables, 2026). Since

flexible networks can also fit the noise in the training data, we keep the network small and add weight decay, which can improve generalisation (Krogh & Hertz, 1992, p. 950). The 13 predictors are rescaled to [0, 1], as inputs on similar scales help the weights learn at a similar rate (LeCun et al., 1998, Sect. 4.3). The minimum and maximum come from the training set only, so that no information from the test set leaks into the model (Kapoor & Narayanan, 2023, pp. 1-3).

Tuning and Threshold Selection

We tune the number of hidden neurons (1, 2, 3 or 5) and the weight decay (0.01, 0.1, 0.5 or 1) with the caret package (Kuhn, 2008). As for the random forest, we use 10-fold cross-validation in which all programmes of a country stay in the same fold, because randomly split folds give over-optimistic error estimates and favour overly complex models when observations are dependent (Roberts et al., 2017, p. 914). Models are compared by balanced accuracy, the mean of sensitivity and specificity, because plain accuracy gives an optimistic estimate for a biased classifier on imbalanced data (Brodersen et al., 2010, p. 3121).

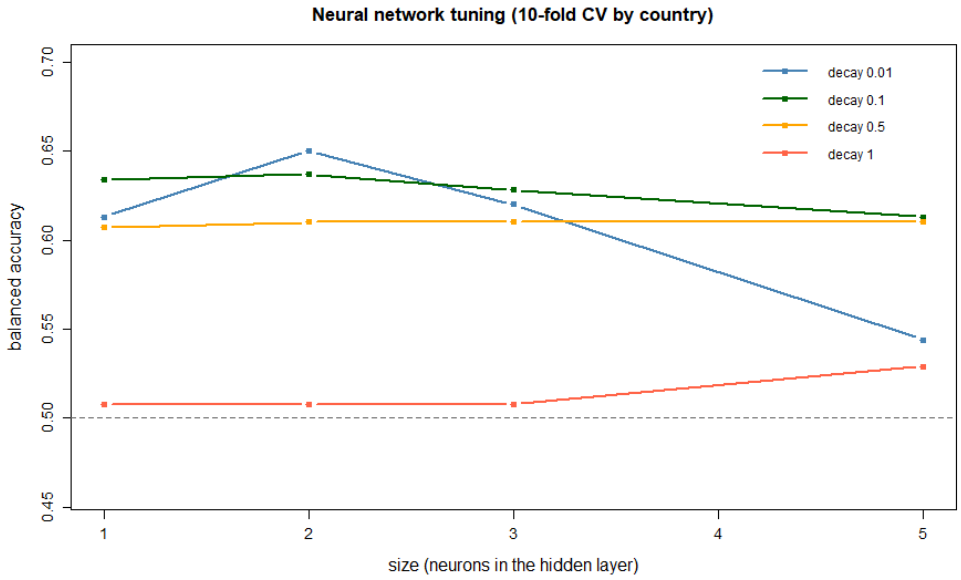


Figure 1: Cross-validated balanced accuracy by number of hidden neurons and weight decay (10-fold CV by country, 145 training programmes; dashed line = 0.500).

The best network has two hidden neurons and a decay of 0.010, with a cross-validated balanced accuracy of 0.650 (Figure 1). With the same light decay, five neurons only reach 0.544, while a strong decay of 1 keeps every size between 0.510 and 0.530, close to the 0.500 of a model that always predicts the same class. With 145 training programmes, a small and lightly regularised network therefore works best. We then choose the classification threshold that maximises balanced accuracy on the out-of-fold predictions, so that the test set plays no role in model selection. This gives a threshold of 0.440 and an out-of-fold balanced accuracy of 0.684.

Test-Set Results and Discussion

Table 5 compares the final network (13 inputs, 2 hidden neurons, 31 weights) with the group's other models on the 28 test programmes approved between 2014 and 2016. The AUC is the probability that a model ranks a randomly chosen interrupted programme above a randomly chosen completed one, so 0.500 corresponds to random guessing (Fawcett, 2006, p. 867).

Table 5: Test-set performance of all models (28 programmes, 2014–2016).

Model	Accuracy	Sensitivity	Specificity	Bal. acc.	AUC
Neural network	0.500	0.294	0.818	0.556	0.567
Random forest (tuned)	0.500	0.471	0.545	0.508	0.586
Logistic regression	0.536	0.412	0.727	0.570	0.684
Baseline (always interrupted)	0.607	1.000	0.000	0.500	0.500

The network beats the random forest on balanced accuracy (0.556 vs 0.508) but not on AUC (0.567 vs 0.586), and both reach an accuracy of only 0.500, below the 0.607 of predicting an interruption for every programme. The two models also make different errors: the network rarely predicts an interruption (sensitivity 0.294, specificity 0.818), while the forest is more balanced (0.471 and 0.545). The network's result is also fragile. With the default threshold of 0.50 it scores slightly higher on the test set (0.572), and refitting it with 20 different random seeds gives test balanced accuracies between 0.436 and 0.559 (median 0.553). This is expected, since the starting weights are drawn randomly (Wüthrich & Merz, 2023, p. 284) and the fitted network is not unique.

We thus find no reliable winner between the two models. With 17 interrupted and 11 completed programmes in the test set, one additional correct prediction moves balanced accuracy by 0.030 to 0.050, as much as the gap between them. The network's balanced accuracy also falls from 0.684 out of fold to 0.556 on the test set, a period in which the share of interrupted programmes rises from 45% to 61%. These results are in line with evidence that tree-based models still outperform deep learning on medium-sized tabular datasets of around 10,000 observations (Grinsztajn et al., 2022, p. 507), far more than our 145. Interpretability points the same way: for high-stakes decisions, Rudin (2019) argues for inherently interpretable models over black boxes that are too complicated for humans to comprehend, and the network's 31 weights have no direct economic meaning. Since it brings no clear accuracy gain over the random forest, nor over logistic regression, which has the highest balanced accuracy and AUC in Table 5, we would not use the neural network to flag programmes at risk of interruption.

REMARKS

Using 13 predictors available at program approval, we fitted a random forest to predict IMF program interruptions. Country-grouped cross-validation selected 250 trees and three candidate predictors per split. The final model reached a pooled development out-of-fold balanced accuracy of 0.630, but only 0.508 on the 28 later test programs at the selected threshold of 0.51. Scheduled reviews, extended facilities and loan access had the highest estimated permutation importance, although these rankings do not establish causal effects.

The results are limited by the small test sample, the change in interruption frequency between development and testing, and the use of development validation results for both model and threshold selection. Test classifications are also sensitive to the threshold: moving from 0.50 to 0.51 changes two predictions. Furthermore, scheduled reviews enter the outcome definition, giving their predictive importance a partly mechanical basis.

Future work could expand the sample, assess performance over additional chronological splits, and investigate richer approval-time indicators of political commitment and institutional capacity. Preprocessing and model selection should be confined to the training portions of validation splits. Until performance is shown to be stable across further samples, this forest is best regarded as an exploratory model rather than a dependable stand-alone signal for sovereign-risk decisions.

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Resource	Use	Affected Section
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ChatGPT	Grammar/spelling correction and rephrasing Help by Correction/Designing code	Entire paper + Code

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